

Transparent observation handling in a study analytics demonstration

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Format: Interactive demonstration

ABSTRACT: Learning analytics depends on distinctions that can disappear before an educator sees a dashboard: a measured zero versus an absent score, an unfinished retry versus a completed attempt, and a withdrawn participant versus an active record. This interactive demonstration presents VertexED's observation review workflow for a study sequence of attempt, review, and retry. Six constructed session records make each distinction inspectable without using learner data. Attendees replace a measured zero with a missing value, restore the observation, withdraw an example participant, and inspect the recomputed aggregate. The same repository analytics functions validate records and generate the downloadable output. Review actions connect rejected or incomplete observations to source repair and follow-up questions, while preserving uncertainty about learning. Aggregate exports omit participant identifiers and session rows; the demonstration also explains why aggregation alone cannot guarantee anonymity in small cohorts. The contribution is a reproducible tool demonstration of transparent observation handling, accompanied by a runnable example and captioned walkthrough. Software checks assess data handling and interaction behaviour. They provide no evidence of educational efficacy, institutional adoption, or causal learning gains. The session invites learning analytics practitioners to examine which missing observations require action before interpreting a study summary or planning an educational intervention.

Keywords: learning analytics; observation quality; missing data; consent; study review

1 DEMONSTRATION MOVIE

Captioned walkthrough of the executed example, 110 seconds, no audio.

<https://github.com/vertex-studyAI/vertexED.ai/blob/conference/20261007-pilot-observation-integrity/docs/demos/lak2027/walkthrough.mp4>

2 DEMONSTRATION SCOPE

The local tool exposes source validation, incomplete-session interpretation, withdrawal and aggregate export. Every record is constructed. No learner study or intervention outcome is reported. At the demonstration, attendees can question the decisions and inspect the exact source and output.

3 USE OF ARTIFICIAL INTELLIGENCE

ChatGPT assisted with drafting, implementation and verification of this demonstration package. Generated code and text are accompanied by executed software evidence. The presenting author remains responsible for reviewing the contribution, attribution, limitations and final submission.